Progress Check

Natural Disaster Severity Prediction

Group 5

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May 21, 2026

GitHub: https://github.com/skyhong2002/disaster-severity-prediction

Task and Data

Predict each region's drought severity for the **next five weeks** from the 91-day test window.

Property	Value
Regions	2,248
Train rows	12.3M
Train days / region	5,480
Test days / region	91
Meteorological features	14

Metric: **MAE**, lower is better.

Main Data Observation

The target `score` is weekly, but the meteorological data is daily.

- About **6/7 daily score values are NaN**, so labels are sparse relative to features.
- The core modeling issue is aligning **daily weather signals** with **weekly drought severity**.
- Test data has no `score`, but score history is still potentially useful for forecasting.

Current Pipeline

We implemented a reproducible **Direct Forecasting Ensemble**.

1. Build temporal features from weather, calendar, and region statistics.
2. Incorporate newly introduced domain-specific Drought and Dryness Indices.
3. Train direct horizon models for both **LightGBM** and **XGBoost**.
4. Blend the predictions (Ensemble) to generate a robust Kaggle submission.

Feature Engineering

Current model input has a diverse set of features.

- **Weather dynamics**: lags, rolling mean/std, exponentially weighted means.
- **Time and location**: calendar encodings and region-level historical score statistics.
- **Domain Features**: Drought Index (temp/precip) and Dryness Index (temp range * max temp) to capture explicit drought conditions.
- *Note*: Historical score lags were completely removed due to severe train-test noise and discrepancy.

Direct Multi-Step Forecasting

Train five independent models for `week1`, `week2`, `week3`, `week4`, and `week5`. Each model predicts one future week directly.

Ensemble Strategy

Train both **LightGBM** (leaf-wise growth) and **XGBoost** (depth-wise growth) on the exact same chronological features. The final submission is a 50/50 blend of both, capturing maximum model diversity.

Local Validation Results (Chronological Split)

We fixed a critical **Time-Leakage** bug by switching to a strict chronological holdout (last 20% of time).

Horizon	LGBM MAE	XGB MAE
Week 1	0.6737	0.7000
Week 2	0.6837	0.7055
Week 3	0.6945	0.7160
Week 4	0.7049	0.7239
Week 5	0.7141	0.7298
Average	**0.6942**	**0.7150**

Kaggle Public Result

Public leaderboard uses about **40%** of the test data.

Method	Public MAE
Baseline 3	0.8056
Team 20	0.8062
Team 5 (v0 Leaky)	**0.8094**
Team 5 (Final Ensemble)	**0.8232**
Baseline 2	0.8623
Baseline 1	0.9117

The ensemble successfully improves upon the single un-leaked LGBM model (0.8299).

Engineering Progress

The codebase is prepared for future model iterations.

- Main scripts: `src/train.py`, `src/predict.py`, `src/features.py`.
- Each training run saves `config.json`, `metrics.json`, and submission metadata under `experiments/<run_id>/`.
- Large model files and generated submissions are kept out of git, while small metadata files can support report updates.

This avoids overwriting the current baseline when we test new algorithms.

Key Discoveries & Challenges

- **Data Leakage Solved**: Initial region-based splits caused artificial local MAE (0.21) vs LB MAE (0.80). Chronological splitting aligned them.
- **Train-Test Discrepancy**: Attempting to reconstruct missing test labels created ≈ 0.74 MAE noise, poisoning our lag features.
- Removing label-dependent features entirely yielded a much cleaner, memory-efficient pipeline.

Next Steps

Before the final deadline, we plan to:

- Tune XGBoost hyperparameters formally to close the gap with LightGBM.
- Try more complex rolling windows and feature selection techniques.
- Explore sequence-based models like Temporal Fusion Transformer.
- Finish documenting the methodology in the IEEE LaTeX report.

Summary

Completed so far:

- Resolved critical Train-Test Leakage and Feature Discrepancies.
- Transitioned to Pure Direct Forecasting with Domain Features (Drought/Dryness Index).
- Implemented and evaluated XGBoost + LightGBM Ensemble.
- Kaggle Public MAE: **0.8232** (clean, reproducible, un-leaked).

Main focus next: hyperparameter tuning and model refinement.

Thank you